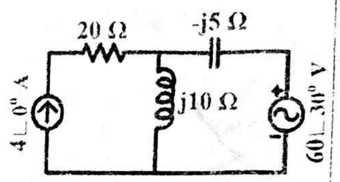
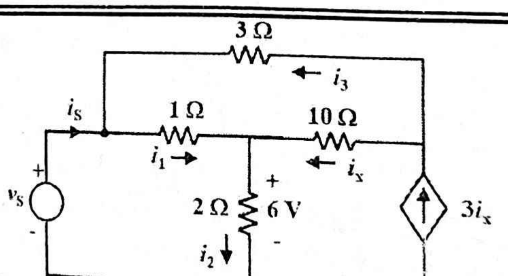
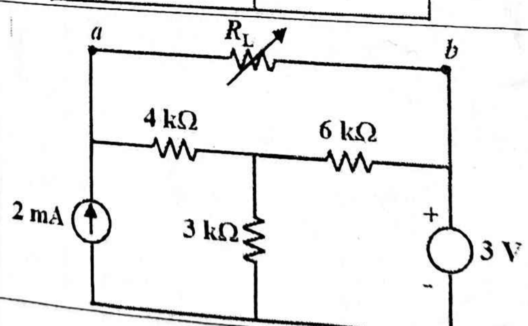
DEPARTMENT OF ELECTRICAL ENGINEERING
IIT (ISM) DHANBAD

Examination: **I semester – B.Tech. (A,B,C,D)** Semester: Monsoon Session: 2022-23
Subject: **Basics of Electrical Engineering (EEI101)** Time: 3 Hours Max Marks: 100

Note: 1. There are total **4 questions**, and all parts of a question should be answered at one place.
2. Make suitable assumptions as necessary and clearly state them.

<table>
<tr><th colspan="2">Q. No. (Part)</th><th>Questions</th><th>Marks</th></tr>
<tr><td rowspan="4">1.</td><td>(1A)</td><td>Answer (in not more than 3-4 lines) the following:
(a) What do you understand by the term 'lead' and 'lag' in relation to alternating quantities? Explain with waveform.
(b) What are apparent power and complex power? State the difference between the both.
(c) What is meant by frequency in an alternating waveform? What is the significance of the quality factor in resonance circuits (don't state formula)?</td><td>[2+2+2]</td></tr>
<tr><td>(1B)</td><td>In the circuit shown in the figure, find the following:
(a) Average power supplied by voltage source.
(b) Average power supplied by the current source.
(c) Average power absorbed by the resistor and the capacitor.

</td><td>[2+2+2]</td></tr>
<tr><td>(1C)</td><td>When a voltage of 120 V at 50 Hz is applied to coil-A, the current drawn is 6 A, and the average power consumed is 160 W. When the same voltage is applied to coil-B, the current drawn is 12 A, and the average power consumed is 420 W.
(a) Find the resistance and the reactance of coil-A and coil-B.
(b) Obtain the current phasor of the current drawn when 100 V is applied across the two coils, coil-A and coil-B connected in series. Find the resultant power factor. Consider the supply voltage as reference phasor.</td><td>[4+4]</td></tr>
<tr><td>(1D)</td><td>Consider a 'series RLC circuit' fed by a variable frequency voltage source. Derive the expression for the frequencies, at which the circuit current is exactly half as much as the current drawn during the series resonance.</td><td>[5]</td></tr>
<tr><td rowspan="2">2.</td><td>(2A)</td><td>For the network shown aside, find the branch currents i_1, i_2, i_x, i_3, and the ratio v_s/i_s.

</td><td>[7]</td></tr>
<tr><td>(2B)</td><td>For the circuit shown aside, find the power delivered to the load resistance (R_L) under maximum power transfer condition.

</td><td>[5]</td></tr>
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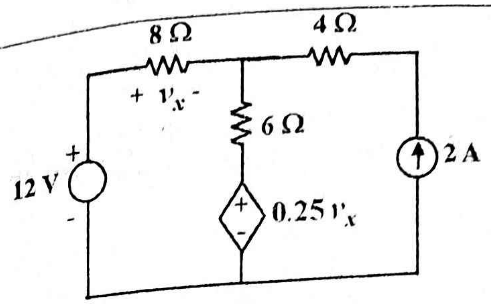
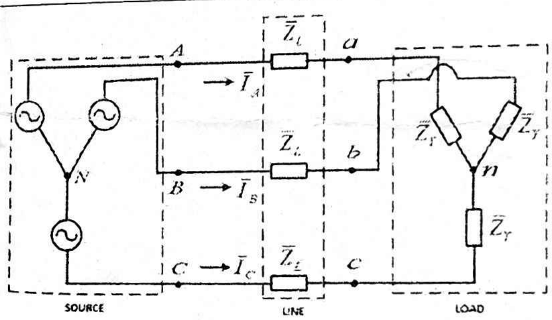
	(2C)	A DC network with a dependent voltage source is shown in the figure aside. Find: **(a)** Magnitude of the dependent voltage source using the superposition theorem. **(b)** Current flowing through 6 Ω resistor. **(c)** Power delivered or absorbed by the dependent voltage source.  	[4+2+1]
	(2D)	**(a)** State and prove the maximum power transfer theorem for a DC network. **(b)** Prove that the Thevenin's equivalent of any network can be converted into Norton's equivalent by using the relationships $I_N = V_{TH}/R_{TH}$ and $R_N = R_{TH}$. Here, I_N and R_N are the Norton's equivalent current and equivalent resistance, respectively. V_{TH} and R_{TH} are the Thevenin's equivalent voltage and resistance of the network, respectively.	[4+2]
3.	(3A)	In a balanced 3-phase system, the phase currents lag the respective phase voltages. The load power is measured by using the two-wattmeter method. The two wattmeters read 108.6 kW and 41.4 kW. Calculate the active power, reactive power, apparent power, complex power drawn by the load, and the load power factor. If the power factor of the load is improved to unity keeping the active power the same, what would be the reading of the two wattmeters used to measure the power of the load?	[7]
	(3B)	Prove that the instantaneous power in a balanced three-phase AC system is a constant.	[5]
	(3C)	In the three-phase system shown in the figure, the phase sequence is ABC, voltage $\bar{V}_{AB} = 340\angle 0°$ V, $\bar{Z}_L = 2.5\angle 30°\ \Omega$ and $\bar{Z}_Y = 20\angle 60°\ \Omega$. Calculate the currents $\bar{I}_A$, $\bar{I}_B$, $\bar{I}_C$, voltages $\bar{V}_{BC}$, $\bar{V}_{CA}$, $\bar{V}_{ab}$, $\bar{V}_{bc}$, $\bar{V}_{ca}$, active and reactive powers supplied by the source, consumed by the load, and lost in the line, magnitude of phase and line-to-line voltage at the load terminal, resultant per-phase impedance and the resultant power factor of the system.  	[13]
4.	(4A)	The voltage applied to the stator of a three-phase, 4 pole, and induction motor has a frequency of 50 Hz. The frequency of the emf induced in the rotor is 2 Hz. Find: **(a)** the slip in percentage; **(b)** rotor speed in rpm; and **(c)** the synchronous speed of the induction motor in rpm.	[2+2+2]
	(4B)	For a 5 kVA, 50 Hz, 230/115 V, single-phase transformer, the iron loss is found as 100 W at no load while rated voltage is applied at the high-voltage side. At no load, the apparent power drawn is 400 VA. Find **(a)** the iron loss component of the no load current, **(b)** magnetizing component of the no load current, **(c)** total no load current, **(d)** no load power factor, and **(e)** rated full load current of the secondary winding.	[5×2]
	(4C)	A 3-phase, 460 V, 60 Hz, four-pole induction motor has a slip of 5%. Determine **(a)** the speed of the rotating magnetic field and **(b)** the frequency of the rotor current in Hz.	[2+2]
	(4D)	Discuss in brief the operating principle of a single-phase transformer with a neat circuit diagram. Derive the expression for induced emf (RMS value) in a transformer winding considering sinusoidal voltage is applied on the primary winding.	[3+2]